HAO "TROY" YU

Address: Rego Park, NY **Portfolio Website:** troyyu527.github.io/ReactWebsite
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Software Engineer | BIM & Automation Developer

PROFESSIONAL SUMMARY

10+ years in the AEC industry, including 6+ years leading BIM initiatives and 4–5 years of hands-on software development alongside that work. Designs and builds production software end to end — C#/.NET desktop applications and plugins, TypeScript web apps and cloud services, and API integrations that connect engineering tools, models and data — with clean architecture and code review. Led a firm-wide CAD-to-BIM transition and set BIM standards and digital-delivery workflows at AECOM. Turns manual remodeling and data re-entry into governed, parameter-driven automation, and builds multi-agent AI and RAG applications that check building models against code with cited evidence. Bridges engineering, BIM and software teams.

AREAS OF EXPERTISE

Software Design & Architecture ~API Development & Integration ~Desktop & Web Applications
Multi-Agent AI & RAG Systems ~Data Translation & Automation Pipelines ~Cloud Services & Web APIs
BIM/CAD Automation ~Model QA/QC & Validation ~Digital Delivery & Standards

TECHNICAL ACUMEN

Languages: C# | TypeScript | JavaScript | Python | SQL
Frameworks: .NET | WPF | MVVM | React | Vite | Node.js | REST APIs | WebSockets | Three.js | D3.js
Engineering APIs: Revit | AutoCAD / Civil 3D | Tekla Open API | IFC (xBIM) | Bentley ProjectWise | Dynamo
Data & Cloud: SQL | SQLite | MySQL | Cloudflare Workers | D1 | Durable Objects | Vectorize | ACC / BIM 360
Delivery: Git | GitHub Actions | Azure DevOps | Agile Scrum
AI: LLMs | RAG | LangGraph | Multi-Agent Workflows | MCP | Workers AI | Ollama | AI-Assisted Development

RELEVANT EXPERIENCE

BlueScope Buildings North America, Inc. • Kansas City, MO, USA
Software Engineer (Contract via Innova Solutions) 01/2026 – 06/2026

- **Built plugins and engineering applications for BlueScope's in-house, model-driven design platform in C#, .NET and WPF/MVVM on the Tekla Open API — UIs whose ViewModels enforce input-format and structural-rule validation before data reaches the model.**
- **Owned the UI side of migrating legacy plugins onto the next-generation platform: redesigned validation and screens, refactored ViewModels and bindings into reusable components, and improved maintainability and long-term support.**
- **Automated steel-building design and detailing steps through API-driven tooling, reducing manual effort.**
- **Trained through BlueScope's AI-development program and used Copilot and LLMs daily in production; built a library of reusable prompts and skills for recurring coding tasks, which let legacy plugins be migrated to refactored code and redesigned UIs much faster than by hand.**
- **Designed and implemented assigned features end to end, using LLMs as a productivity aid for code review, research, debugging and documentation, with every suggested change reviewed before merge.**
- **Took part in architecture discussions, solution design reviews and sprint planning in an Agile Scrum team (Azure DevOps).**

AECOM • New York, NY, USA
BIM Supervisor – Digital Solutions 01/2024 – 12/2025

- **Led the development and rollout of BIM standards, protocols and digital-delivery workflows for multidisciplinary Civil, Structural and BIM teams on NYC Economic Development Corporation waterfront infrastructure (United Nations Waterfront Esplanade, ~$120–153M; Seaport Coastal Resiliency, up to $229M).**
- **Ran model validation, QA/QC and data management against project standards and client requirements; deployed project environments on Autodesk Construction Cloud and governed model exchange with partners.**
- **Key Automation & Software Contributions:**
  - **Eliminated repeat remodeling on the UN Esplanade, where the alignment kept changing: secured budget hours, aligned Civil, Structural, BIM and the PM, and architected a parameter-driven Civil 3D → Excel → Revit pipeline — a C# command exports alignment data, design parameters live as structured metadata, and Dynamo rebuilds the model. A design change became a data update instead of a remodel.**
  - **Wrapped the C/C++ Bentley ProjectWise API in a reusable C# library and built a desktop file explorer so the team could read project files without the full check-out workflow.**
  - **Developed Revit API add-ins and C# libraries for parameter extraction, element filtering and model analysis; proved approaches in C#, then handed them over as Dynamo graphs the BIM team could maintain.**
  - **Automated data extraction and translation with Dynamo, Python, SQL/SQLite and Excel, making deliverables consistent and reducing manual processing.**

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Urbantech Consulting Engineering PC • New York, NY, USA

BIM Engineer 10/2014 – 12/2022

- **Led the firm's transition from CAD to BIM: established BIM standards, modeling practices, family and parameter libraries, coordination workflows and digital information management; led a team of ~5 engineers implementing BIM and in-house extensions.**
- **Delivered model-to-construction BIM for >$6B of New York infrastructure (LaGuardia Terminal B ~$4B, Goethals Bridge $1.5B, Kosciuszko Bridge $555M, Unionport Bridge $232M) for PANYNJ, NYSDOT, NYC DOT and MTA contractors.**
- **Key Automation & Software Development Contributions:**
  - **Developed Revit, AutoCAD and Civil 3D plugins in C# using custom metadata structures to automate data extraction, model management and CAD↔BIM interoperability — an estimated 10–15% productivity gain for daily users.**
  - **Ran an end-to-end data pipeline (ETL) combining GIS models, LiDAR point-cloud scans and design data into BIM models; reduced point-cloud noise in Python to align scans with designs.**
  - **Built a Python rebar bill-of-materials app with automatic validation for reinforcing-steel design errors, and a C# Autodesk license manager for real-time seat tracking.**

SELECTED SOFTWARE PROJECTS

BIM AI Agent (live) / ***TypeScript, LangGraph.js, RAG, Cloudflare Workers AI, Vectorize, D1, Three.js, web-ifc***

- **Multi-agent QA pipeline that checks IFC building models against the NYC building codes: a Planner picks the code topics for each model, then a Researcher, Rule Author, Checker and Reviewer turn the code text into checkable rules with verbatim citations and graded severities; clashes, fire-door ratings and model data are checked deterministically.**
- **Hybrid RAG over 168 official NYC code PDFs (7,045 chunks): vector and full-text search fused and reranked, every citation with section, page and a PDF preview. A live AI assistant routes each question to code search, the findings or the model data, answers with evidence and drives the 3D view; it advises on fixes and never edits the model.**
- **Browser app (React, Three.js, web-ifc) on a Cloudflare Worker with Durable Objects streaming the agent steps; a daily AI budget guard, per-session limits and cached searches keep it on the free plan, and the same code runs locally on Ollama.**

Engineering Model QA / ***C#, .NET, WPF/MVVM, xBIM, IFC4***

- **Rule-based IFC model-checking workbench: governed JSON rule profiles (required property, unique value, naming pattern, material, storey assignment) produce deterministic, explainable findings with guidance.**
- **Revision Compare classifies every finding as Resolved, Persistent or New across model versions by matching IFC GlobalIds and stable condition keys; each rule returns failed, passed or not evaluated, so missing data never produces a false failure.**

IFC Viewer / ***C#, WPF/MVVM, xBIM, DirectX 11***

- **Desktop IFC2x3/IFC4 viewer: 3D navigation, picking with highlight, isolate/hide, a properties panel and a spatial tree linked both ways with the 3D view, plus search by Revit Element ID.**
- **Opens models of several hundred MB on a background thread by switching large files to on-disk storage (a 385 MB, 56,000-element model loads without running out of memory).**

AI Career Assistant (live) / ***TypeScript, LangChain.js, Cloudflare Workers, Workers AI, Vectorize, React***

- **Production RAG assistant over a curated knowledge base of my career: questions are rewritten with the conversation, searched in a vector index and answered from the retrieved facts only, with streamed answers, source chips and project media; input validation, a CORS allow-list and per-IP rate limiting; embedded in my portfolio.**

UNO Multiplayer Game (live) / ***TypeScript, React, Zustand, Cloudflare Workers, D1, Durable Objects***

- **Real-time multiplayer on Durable Objects and WebSockets with a single-writer authority model; presence tracking hands an abandoned seat to the AI, players reconnect into their seat, and newcomers claim AI seats mid-game through server-serialized claims.**
- **Worker backend with PBKDF2 password hashing and JWT sessions and saves in D1; game rules as a pure, immutable TypeScript engine driving a heuristic AI.**

Hide & Seek Pathfinding Visualizer (live) / ***React, Vite, JavaScript***

- **Animates six graph-search algorithms (Dijkstra, A*, BFS, DFS, greedy best-first, bidirectional) on a weighted grid with terrain cost and generated mazes; a race mode runs all six on one map and compares path cost against cells explored.**

Data Structures & Algorithms Visualization (live) / ***D3.js, JavaScript, SVG***

- **Step-by-step D3/SVG animations of eight data structures and six sorting algorithms plus a 3D force-directed graph explorer, built on a registry-driven module architecture as a zero-build static site.**

EDUCATION

Master of Science in Civil Engineering and Engineering Mechanics • Columbia University • 2014
Bachelor of Science in Harbor and River Engineering • National Taiwan Ocean University • 2010

MSV (Mechanics Sound Vibration) Lab, National Taiwan Ocean University • Keelung, Taiwan

Research Assistant – Computational Analysis & Modeling 2009 – 2010

- **Built MATLAB numerical models for a National Science Council civil engineering research project, applying finite element, boundary element and finite difference methods to engineering mechanics problems.**
- **Collected data, ran model computations and helped develop mathematical formulations for papers published in SCI journals (e.g., Engineering Analysis with Boundary Elements).**